VII. Conclusion

This thesis proposed EPFD-Aware Beam Reassociation for Service Recovery (EABR), a beam reassociation framework for EPFD-constrained multi-beam LEO satellite systems under NGSO–GSO coexistence. EABR exploits predictable satellite motion, beam footprints, and coverage overlap among neighboring LEO satellites to support users affected by beam shutdown. The framework consists of Safe-Beam Reassociation for Helper Users (SBR), EPFD-constrained power allocation for helper recovery beams, and Helper-Beam Reassociation for Closed-Beam Users (HBR).

The simulation results show that EABR keeps the aggregate EPFD below the required limit during the critical period. The helper availability analysis under different constellation densities also shows that helper satellite candidates are generally available under the evaluated constellation densities. Compared with beam shutdown only, PC + Tilt, and Only HBR, EABR achieves higher average user satisfaction under different user loads. The results also show that SBR increases the number of recovered closed-beam users, especially around the worst EPFD slot, and the satisfaction CDF confirms that EABR reduces the proportion of low-satisfaction users. The online computational overhead evaluation further shows that the average execution time of EABR remains below the 1-s time-slot duration under all evaluated user loads.

Overall, EABR provides a practical service recovery approach for multi-beam LEO satellite systems under the GS EPFD constraint. By using precomputed beam relation information and neighboring helper satellites, the proposed framework improves service continuity with a practical time-slot-based decision process. Therefore, EABR achieves a better balance among EPFD compliance, user satisfaction, and online computational overhead.

References

[1] M. Giordani and M. Zorzi, "Non-terrestrial networks in the 6g era: Challenges and opportunities," *IEEE Network*, vol. 35, no. 2, pp. 244–251, 2021.

[2] X. Lin, S. Cioni, G. Charbit, R. Chuberre, S. Hellsten, and J.-F. Boutillon, "On the path to 6g: Embracing the next wave of low earth orbit satellite access," *IEEE Communications Magazine*, vol. 59, no. 5, pp. 36–42, 2021.

[3] E. Juan, M. Lauridsen, J. Wigard, and P. Mogensen, "Handover solutions for 5g low-earth orbit satellite networks," *IEEE Access*, vol. 10, pp. 93 309–93 325, 2022.

[4] M. Al-Ansi, J. Querol, M. Alsenwi, E. Lagunas, and S. Chatzinotas, "Enhancing handover performance in LEO satellite networks with multi-connectivity and conditional handover approach," in *Proceedings of the International Conference on Software, Telecommunications and Computer Networks (SoftCOM)*, 2024.

[5] A. Gonzalez-Garrido, J. Querol, H. Wymeersch, and S. Chatzinotas, "Joint communication and navigation from LEO multi-beam satellite," *IEEE Open Journal of the Communications Society*, vol. 6, pp. 4384–4404, 2025.

[6] D. Christopoulos, S. Chatzinotas, J. Krause, and B. Ottersten, "Multi-user detection in multibeam mobile satellite systems: A fair performance evaluation," *IEEE Transactions on Communications*, vol. 61, no. 12, pp. 4963–4974, 2013.

[7] X. Zhu and C. Jiang, "Integrated satellite-terrestrial networks toward 6g: Architectures, applications, and challenges," *IEEE Internet of Things Journal*, vol. 9, no. 1, pp. 437–461, 2022.

[8] O. B. Al-Hraishawi, H. Chaaban, S. Kisseleff, E. Lagunas, and S. Chatzinotas, "A survey on non-geostationary satellite systems: The communication perspective," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 1, pp. 101–132, 2023.

[9] ITU-R, "Radio regulations," International Telecommunication Union, Geneva, Switzerland, 2020, see Article 22 for NGSO–GSO interference limits and EPFD framework.

[10] ——, "Functional description to be used in developing software tools for determining conformity of non-geostationary-satellite orbit fixed-satellite service systems or networks with limits contained in article 22 of the radio regulations," International Telecommunication Union Radiocommunication Sector (ITU-R), Geneva, Switzerland, Recommendation S.1503-3, 2018.

[11] International Telecommunication Union Radiocommunication Sector (ITU-R), "Satellite antenna radiation patterns for non-geostationary orbit satellite antennas operating in the fixed-satellite service below 30 ghz," ITU-R, Geneva, Switzerland, Recommendation S.1528, 2001.

[12] ITU-R, "Reference fss earth-station radiation patterns for use in interference assessment involving non-gso satellites in frequency bands between 10.7 ghz and 30 ghz," ITU-R, Geneva, Switzerland, Recommendation S.1428-1, 2001.

[13] ——, "Calculation of free-space attenuation," International Telecommunication Union Radiocommunication Sector, Geneva, Switzerland, Recommendation P.525.

[14] ——, "Attenuation by atmospheric gases and related effects," International Telecommunication Union Radiocommunication Sector, Geneva, Switzerland, Recommendation P.676.

[15] ETSI, "5g; system architecture for the 5g system (5gs) (3gpp ts 23.501 version 16.14.0 release 16)," European Telecommunications Standards Institute, ETSI TS 123 501 V16.14.0, Sep. 2022, 3GPP TS 23.501 version 16.14.0 Release 16.

[16] C. Braun, A. M. Voicu, L. Simić, and P. Mähönen, "Should we worry about interference in emerging dense NGSO satellite constellations?" in *Proc. IEEE Int. Symp. Dyn. Spectr. Access Netw. (DySPAN)*, 2019, pp. 1–10.

[17] A. Hills, J. M. Peha, and J. Munk, "Feasibility of using beam steering to mitigate ku-band leo-to-geo interference," *IEEE Access*, vol. 10, pp. 74 023–74 032, 2022.

[18] M. Jalali, F. Ortiz, E. Lagunas, S. Kisseleff, L. Emiliani, and S. Chatzinotas, "Joint power and tilt control in satellite constellation for NGSO–GSO interference mitigation," *IEEE Open Journal of Vehicular Technology*, vol. 4, pp. 545–557, 2023.

[19] T. Li, J. Jin, W. Li, Z. Ren, and L. Kuang, "Research on interference avoidance effect of oneweb satellite constellation's progressive pitch strategy," *International Journal of Satellite Communications and Networking*, vol. 39, no. 5, pp. 524–538, 2021.

[20] M. Jalali, E. Lagunas, A. Haqiqatnejad, S. Kisseleff, and S. Chatzinotas, "Downlink beamforming strategies for interference-aware NGSO satellite systems," *IEEE Open Journal of the Communications Society*, vol. 5, pp. 3468–3483, 2024.

[21] G. Cocco, M. Angelone, and A. I. Pérez-Neira, "Co-channel interference cancellation at the user terminal in multibeam satellite systems," in *Proceedings of the IEEE International Conference on Communications (ICC)*, 2014, pp. 3992–3997.

[22] C. E. Shannon, "A mathematical theory of communication," *Bell System Technical Journal*, vol. 27, no. 3, pp. 379–423, 1948.

[23] J. Lei and M. A. Vázquez-Castro, "Multibeam satellite frequency/time duality study and capacity optimization," *Journal of Communications and Networks*, vol. 13, no. 5, pp. 472–480, 2011.

[24] WorldVu Satellites Limited, "Oneweb non-geostationary satellite system," FCC IBFS File No. SAT-LOI-20160428-00041, Technical Attachment, Washington, DC, USA, 2016, oneWeb NGSO system description.

[25] Federal Communications Commission, "Space Exploration Holdings, LLC, Request for Modification of the Authorization for the SpaceX NGSO Satellite System," Order and Authorization and Order on Reconsideration, FCC 21-48, Washington, DC, USA, Apr. 2021.

[26] ——, "Space Station Applications Accepted for Filing," Public Notice, Report No. SAT-01632, Washington, DC, USA, May 2022.

[27] Analytical Graphics, Inc., "Systems tool kit (stk)," https://www.agi.com/products/stk, 2023, online; accessed 21-Jul-2025.

[28] The MathWorks, Inc., "Matlab r2023b," https://www.mathworks.com/products/matlab.html, 2023, online; accessed 21-Jul-2025.

[29] ITU-R, "Interference mitigation techniques and frequency sharing in the bands 37.5–42.5 ghz and 47.2–50.2 ghz between geostationary-satellite fixed-satellite service networks and non-geostationary-satellite fixed-satellite service systems," ITU-R, Geneva, Switzerland, Recommendation S.1655-0, 2003.